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1. Title of the invention: Integrating IoT Systems for Sustainable Agricultural Intelligence for Data-Driven Farming and Agricultural Optimization

2. Field /Area of invention: Smart Agriculture, Precision Farming, Sustainable Technology & Renewable Energy

3. Prior Patents and Publications from literature (provide a table summarizing the prior art):

S.No	Patent/Publication	Methodology	Limitations
1	US20180262571A1 [2017]	Automation to improve farming efficiency and sustainability. AI models analyze data to optimize irrigation, fertilization, and pest control.	User-friendly dashboards for farmers, automated control systems, and integration with supply chains ultimately enhance crop yields, conserve resources, and support data-driven decision-making in agriculture.
2	US10728336B2 [2017]	A user-friendly interface helps farmers make informed decisions remotely to optimize irrigation, fertilization, and pest control.	Providing predictive insights and personalized crop recommendations to improve productivity, resource efficiency, and sustainability in modern farming.
3	US9792557B2 [2017]	To assist in identifying alerts—such as potential crop problems—and suggesting actions to resolve them.	To reduce operational costs, boost crop yields, improve food safety, and minimize environmental impact through data-driven, site-specific farming practices.
4	US20240180062A1 [2020]	By capturing the shape of the field and creating a virtual model, this system allows machinery to plan travel routes precisely, avoiding obstacles and irregularities.	Ensures accurate and efficient field operations, minimizing time and resource waste while boosting overall productivity.
5	US20240116630A1 [2020]	It surpasses traditional manual counting and historical data-based predictions, providing a more accurate and scalable solution for farmers.	The system identifies and sizes fruit, providing an accurate yield estimate.
6	US11985914B2 [2024]	Enables tree propagation in difficult conditions, independent of traditional power grids, supporting reforestation and environmental conservation.	Drones plant tree seeds in remote areas using compressed growth media containers. These containers safeguard the seeds until roots develop, and drones recharge with solar panels.
7	US20240168001A1 [2020]	It utilizes wireless soil condition sensors that can be relocated without digging, and AI estimates soil conditions at previously occupied sensor locations.	System optimizes the use of water, fertilizer, and chemicals, reducing waste and promoting sustainable farming practices
8	US11944047B2 [2024]	Employs a drone to optimize crop irrigation by collecting real-time data on crop conditions and ensuring efficient and precise irrigation management.	Drone surveys the crop area and gathers data to customize the irrigation cycle, and the drone's landing position is calculated based on the pivot's movement

4. Summary and background of the invention (Address the gap / Novelty):

Summary:

This proposed work focuses on Sustainable Agricultural Intelligence, utilizing an ESP32 microcontroller and the ThingSpeak cloud platform to enhance farming productivity through real-time soil monitoring and intelligent decision-making. It combines ESP32 microcontrollers with sensors for soil moisture, pH levels, temperature, and light intensity. The system is used to gather data and send it to the ThingSpeak cloud platform. Once the data reaches specific thresholds, it will switch on or off the irrigation motor. This operation helps farmers conserve water and maintain healthier soil.

The system incorporates ThingSpeak for real-time cloud analytics, enabling remote access and prompt decision-making. Its computerized algorithm replaces manual interpretation, making it more accurate and saving labor. The system is made scalable to accommodate future enhancements such as machine learning and broader application on large farms. It emphasizes ease of use with a farmer-friendly remote control, correcting the design flaws of earlier versions. The project recognizes the challenges of rural connectivity and invites more research into offline or edge computing alternatives. Lastly, by recommending effective evaluation techniques and potential field testing, it aims to bridge the lack of empirical studies and performance comparison in existing systems.

The following are the multiple features and difficulties of automated agriculture systems for sustainable intelligence farming and agricultural optimization:

- i) The system solves agricultural issues such as water shortage, wasteful practices, and global warming.
- ii) It simplifies irrigation decisions utilizing real-time sensor information.
- iii) It provides remote monitoring with cloud analytics and an easy-to-use interface.
- iv) It is also applied in scalable design, applicable for small to medium farms with possible integration of machine learning.
- v) It seeks to equip farmers with technology-based intelligence, lessen manual work, and enhance yield from crops in a sustainable manner.

Background:

Agriculture optimization is characterized by challenges such as water shortages, global warming, and poor farming techniques, resulting in low productivity and wastage of resources. Most farmers have no access to soil data in real-time, which hinders informed decisions. Overuse or underuse of water from traditional irrigation techniques can affect soil health and productivity. With a view to overcoming these difficulties, our integrated IoT system is designed to automate irrigation and optimize soil conditions with the help of real-time sensor inputs and cloud analytics. With the help of ESP32 microcontrollers, soil moisture sensors, and pH sensors, the system guarantees optimal use of resources, less manual intervention, and increased crop production. The project is motivated by a desire to equip farmers with cutting-edge technology to ensure agriculture becomes more sustainable, productive, and data-centric.

A number of IoT-based agricultural systems have been designed over the past few years to automate and monitor farming processes. A number of these prominent systems include:

1. **IoT-Based Automated Irrigation System** (2018): This system employs a microcontroller such as Arduino to automate irrigation based on soil humidity. The system is efficient but does not have advanced capabilities like pH monitoring and cloud storage of data.
2. **Soil Moisture and Temperature Monitoring System** (2020): A next-generation system combines temperature and moisture sensors for controlling irrigation but lacks real-time cloud-based analysis or decision-making on various parameters.
3. **IoT and Smart Farming Technologies** (2021): These technologies are starting to utilize cloud platforms such as ThingSpeak, but are often limited to either moisture or temperature sensors and do not integrate multiple sensors. These current systems miss an integrated system, real-time management, and the ability to make decisions on many factors, such as soil pH and moisture level, resulting in reduced efficiency and optimization of resources.

Research Gap:

The proposed work addresses significant research gaps in intelligent agriculture by proposing an IoT-based irrigation system that integrates multi-sensor monitoring of soil moisture, pH, temperature, and light intensity. The strategy overcomes the shortcomings of existing solutions that lead to less efficiency and resource optimization.

- **Multi-Parameter Soil Analysis:** This solution combines pH, temperature, and light intensity sensors compared to traditional systems based only on moisture, offering a complete evaluation of soil health.
- **Real-Time Cloud-Based Decision-Making:** Unported integration with ThingSpeak facilitates remote monitoring and data analysis, going beyond simple local storage and allowing for wiser agricultural decisions.
- **Automated Irrigation Control:** The system automatically initiates irrigation in accordance with pre-defined sensor thresholds, reducing labor-intensive intervention, saving water, and maximizing resource efficiency.

- **Dual Connectivity Using ESP32:** The ESP32 microcontroller, which is Wi-Fi and Bluetooth enabled, provides flexible and secure data transfer, even in regions with intermittent network connectivity.
- **Modular Scalable Design:** Created to be scalable, the system is expandable for large-scale farms and allows for future developments like predictive analytics through machine learning.
- **Small to Medium Farm Support:** Through affordability and simplicity of deployment, the project covers the requirements of the small and medium farmers, a group that is usually overlooked by enterprise-scale technology.
- **Soil Fertility Through Balanced pH:** The feature of pH monitoring and control covers a vital aspect of plant growth and nutrient intake, which is usually left out of current solutions.
- **Remote Access and Control Interface:** Irrigation control and monitoring of soil conditions from a remote location through a simple interface increase convenience and operational effectiveness.

5. Objective(s) of Invention:

To create a multi-sensor smart IoT system that combines sensors to detect soil moisture, pH, temperature, and light intensity in order to obtain real-time environmental information.

- To apply automated irrigation control utilizing sensor-based thresholds to switch a motorized water pump on and off to limit manual intervention and wastage of water.
- To allow for cloud-based real-time analysis by sending data to the Thingspeak platform via an ESP32 microcontroller for remote access and continuous analysis.
- To enhance soil health and crop yields by monitoring critical soil parameters, ensuring optimal conditions, promoting improved nutrient absorption, and boosting crop yields.
- To create a friendly remote monitoring interface for farmers with a simple web-based dashboard to represent sensor data and enable remote irrigation control.
- To make the system scalable and modular, accommodating future expansions like the addition of machine learning-driven predictive analytics and additional sensor integrations.
- To overcome usability and connectivity challenges for rural installations through considerations of edge computing, offline capability, and cost-effective hardware.
- To stimulate sustainable agricultural practices that promote data-driven agriculture, water conservation, reduction of wastage of resources, and long-term environmental sustainability.

6. Working principle of the invention (in brief):

The proposed system involves continuous collection of real-time agricultural parameters such as soil pH, moisture, temperature, and light intensity using respective sensors. These sensor readings are processed by the ESP32 microcontroller, which uses its built-in Wi-Fi to transmit data to the ThingSpeak cloud for storage, visualization, and analytics. Based on predefined threshold values, the system makes intelligent decisions for irrigation and crop management, enabling automation and reducing manual intervention. The proposed work integrates sensors, microcontrollers, and communication modules to automate and monitor soil conditions and irrigation for monitoring and controlling agriculture.

Components of the System:

- **Adapter (12V) and Power supply:** The 12V adapter provides the main power supply to the overall system, delivering a constant 12V DC output. This original power is then supplied to a power supply module that converts and regulates the input 12V to lower voltages like 5V or 3.3V that will be used by the ESP32 microcontroller and other sensors. This voltage regulation is instrumental in determining the stable and safe operation of all electrical components.
- **ESP32 microcontroller:** The brain of the system is the ESP32 microcontroller, which serves as the processing unit. With Wi-Fi and Bluetooth capabilities and a number of GPIO (General Purpose Input/Output) pins, the ESP32 receives data from sensors connected to it, processes it, and forwards it to the cloud or uses it to drive actuators like motors and relays.
- **Soil moisture sensor:** One of the primary sensors connected to the ESP32 is the soil moisture sensor, which measures the water content in the soil. This data helps determine whether irrigation is needed.
- **pH sensor:** It monitors the acidity or alkalinity of the soil, which is an important parameter for plant health and nutrient absorption.
- **ThingSpeak platform:** For remote monitoring and visualization of data, the system employs the ThingSpeak platform, a

cloud-based IoT analytics service. Real-time sensor data is sent by ESP32 to ThingSpeak, making soil conditions observable and analyzable from anywhere.

- **Relay module:** In addition, a relay module is added to the system, which acts as an electronic switch controlled by the ESP32. The ESP32 can, in turn, energize the relay to switch a motor (in this case, usually a water pump) on or off depending on the readings from the soil moisture sensor. The motor then powers the automated irrigation system, pumping water to the soil only when it is necessary to minimize water wastage and avoid the necessity for manual operation.

7. Description of the invention in detail (Include drawings and/or photographs as needed):

IoT-based architecture depicted in Figure 1 enables accurate monitoring, efficient utilization of resources, and higher crop yields compared to traditional and other IoT-based methods. The system architecture relies upon the ESP32 microcontroller, which acts as the central processing unit. The sensors built into the system are a pH sensor for soil alkalinity or acidity measurement, a capacitive soil moisture sensor for water content determination, a DS18B20 or DHT22 temperature sensor for surrounding temperature, and a BH1750 or an LDR light intensity sensor for measuring sunlight exposure. ESP32 gets data from these sensors and sends it over Wi-Fi to ThingSpeak for cloud storage and visualization. Various phases undertaken in this work are as follows:

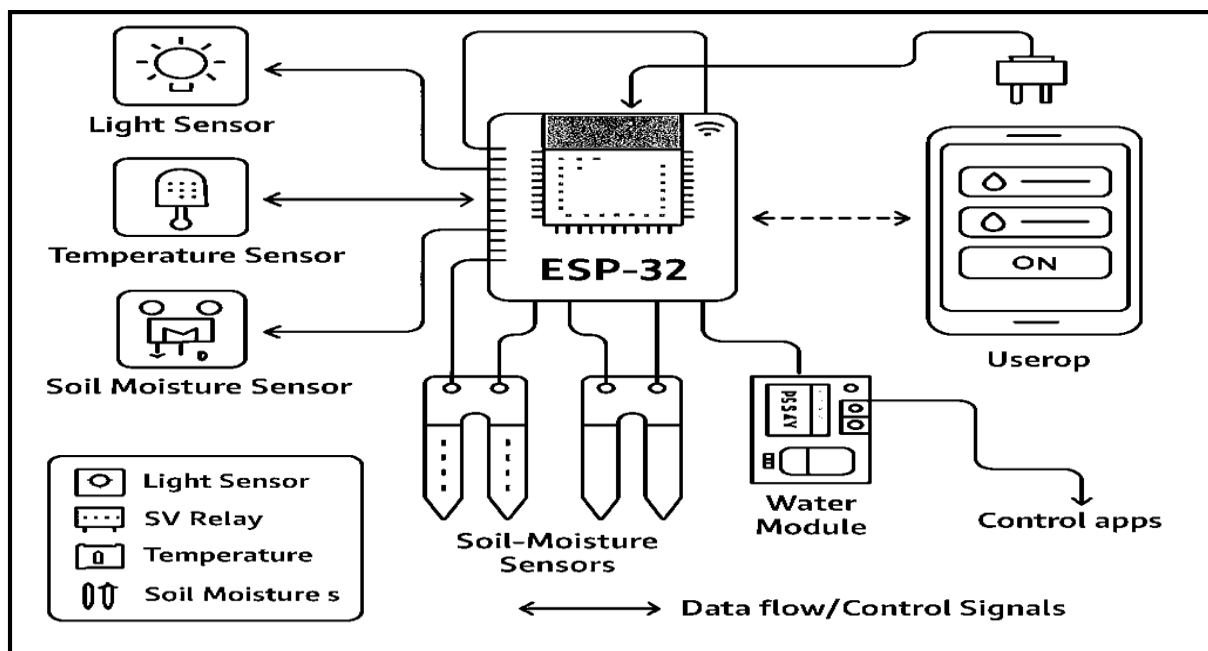


Figure 1: System architecture diagram

1. Sensor Data Acquisition:

The pH sensor, soil moisture sensor, temperature sensor (DHT22), and Light Dependent Resistor (LDR) sensors are used in the agricultural field. These sensors operate in real time to monitor crucial environmental factors such as soil pH level, percentage of soil moisture, room temperature and light intensity. For instance, pH sensors are calibrated by using standard pH buffer solutions of pH 4, 7, and 10, whereas soil moisture sensors are calibrated under dry and saturated soil conditions. Likewise, temperature and light sensors are checked against the standard instruments for reliable readings.

2. Data Processing & Microcontroller Control

ESP32 microcontroller takes analog or digital inputs from each sensor. ADC (Analog to Digital Converter) on ESP32 converts analog signals (from pH, moisture, LDR) into digital signals for processing. Raw values are converted by microcontroller using calibration formulas to useful units ($^{\circ}\text{C}$, %, pH).

3. IoT Connectivity & Cloud Integration:

The application reads sensor values at intervals, filters and calibrates to minimize noise, and uploads processed values onto ThingSpeak with secure API keys. Wi-Fi connection handling and error-correcting mechanisms are also included to ensure smooth data transmission. It uses a deep sleep mode to preserve power between readings.

4. ThingSpeak Configuration and Visualization:

A channel in ThingSpeak is established to hold the sensor readings, with separate fields dedicated for pH, soil moisture, temperature, and light intensity. ThingSpeak offers real-time graphs and past trends in data for simple monitoring. Furthermore, MATLAB analytics in ThingSpeak can be employed to calculate averages, identify anomalies, and initiate alerts when parameters reach specified thresholds, like low soil moisture or unusual pH.

5. Data Logging & Performance Analysis:

Analysis of the gathered readings is done to extract patterns and make knowledge-based decisions. Alerts are set on ThingSpeak to inform users through email or SMS whenever critical values like very low moisture or out-of-range pH are encountered. This serves to enable farmers to make decisions based on quick reactions to field change. Flow diagram of Integrating IoT Systems for Sustainable Agricultural Intelligence is shown in Figure 2.

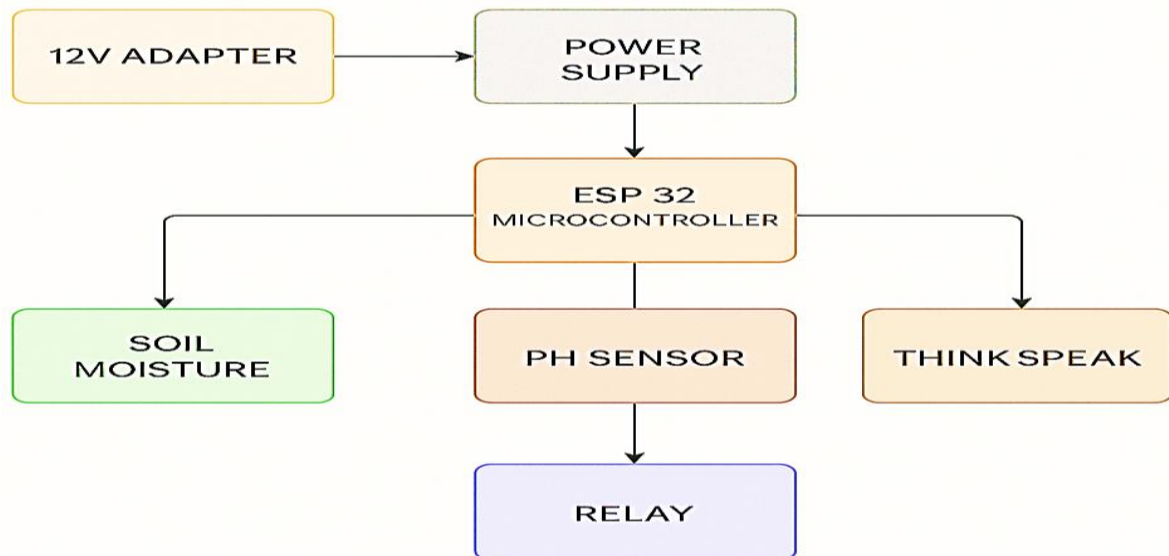


Figure 2: Flow diagram

8. Experimental validation results:

An Experiment was conducted on an agricultural field about 200 m² for a total of 30 consecutive days. The system of smart agriculture employed an ESP32 microcontroller that interfaced with sensors for soil moisture, temperature, intensity of light, and pH of the soil. Sensor readings were transferred to ThingSpeak every 10 minutes. The irrigation system utilized a 12 V DC pump and drip irrigation network, regulated through a relay based on decision-making logic coded in the ESP32. Figure 3 represents a use case diagram of the proposed system.

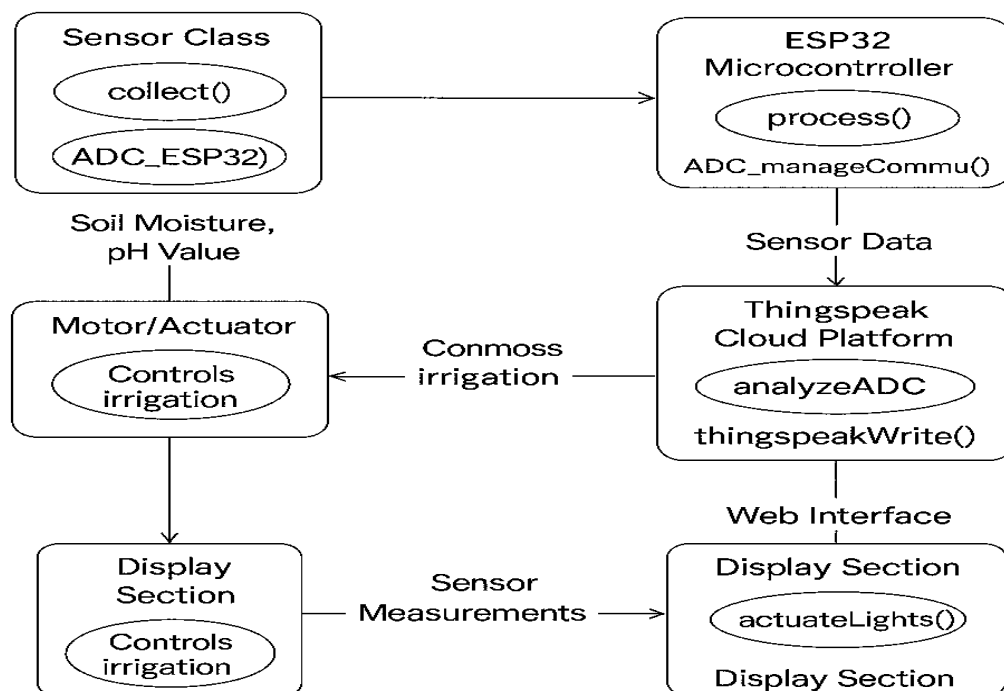


Figure 3: Use Case Diagram

During the experiment as shown in Figure 4, the mean measured values were: soil pH ranged around 8.0 signifies slightly alkaline soil conditions. The soil moisture level is about 2% signifies extremely dry conditions for irrigation. The LDR shows the light intensity at a mean value of 37 units.

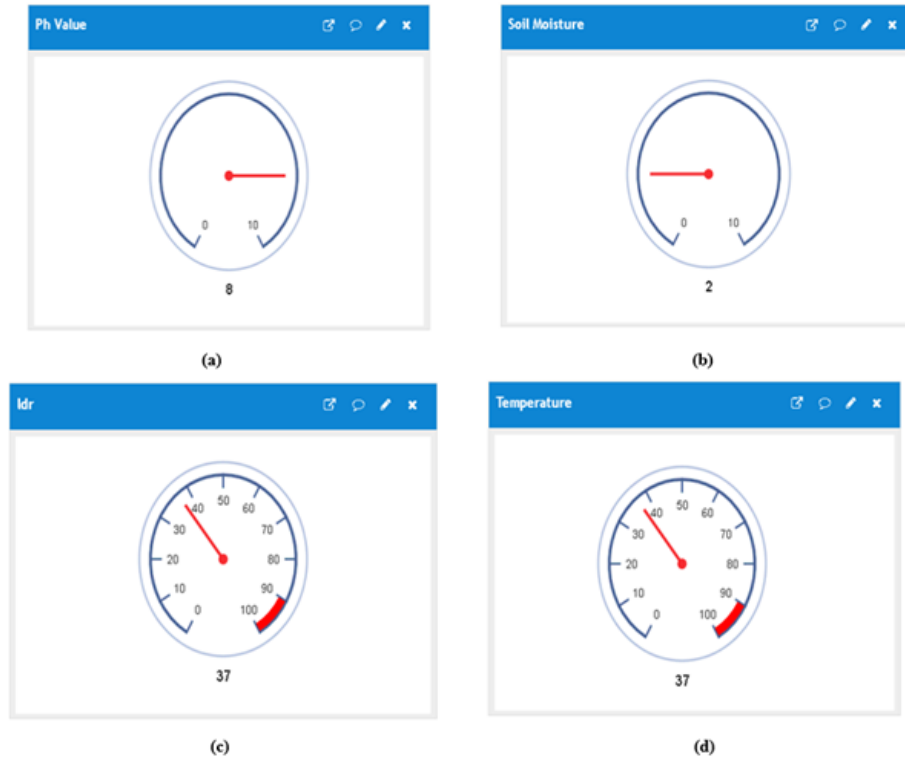


Figure 4: Parameter values: a) pH values b) Soil moisture c) light intensity using LDR d) Temperature

The temperature readings is at an average of 37°C, which signifies high temperature readings, most likely representing sensor locations in direct sunlight or a test case under controlled conditions. Despite these extreme conditions, the system under consideration continuously monitored all parameters and initiated irrigation every time soil moisture fell below the set rate, ensuring better management of soil conditions.

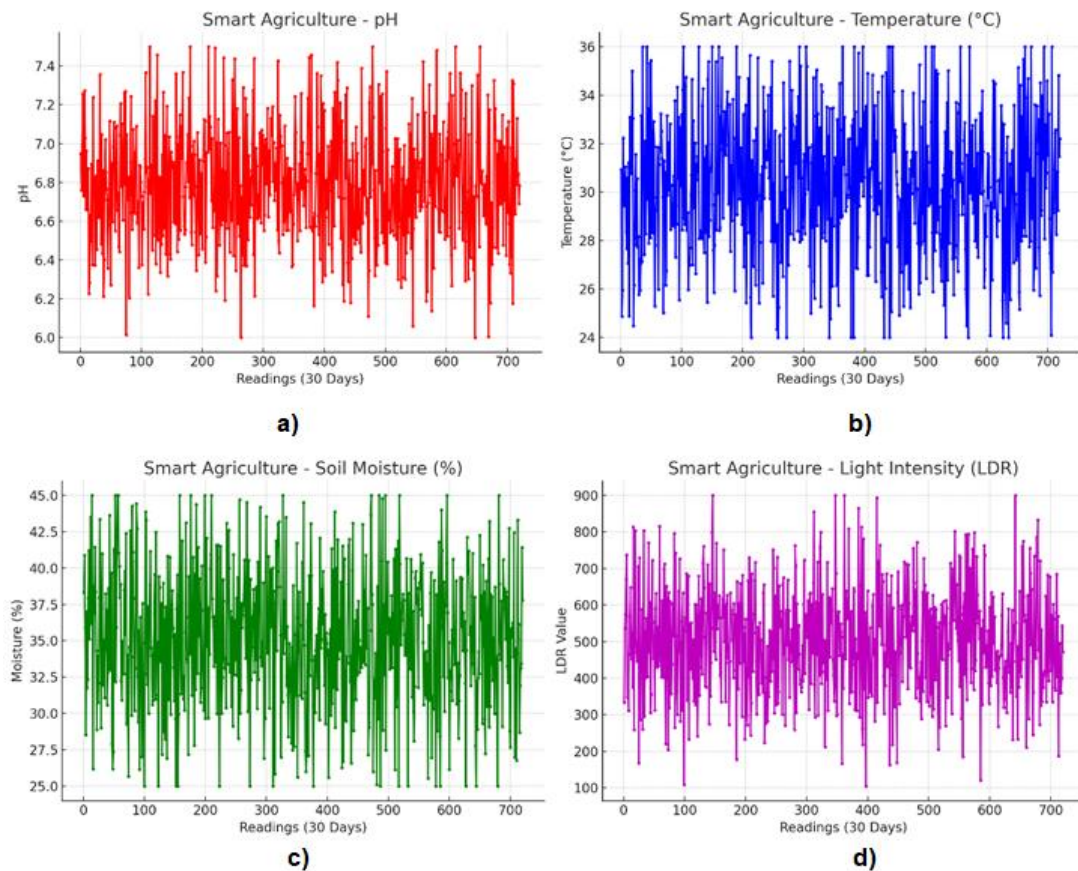


Figure 5: Parameter analysis for 30 days using ThinkSpeak Graph: a) pH values b) Temperature c) Soil moisture d) light intensity

Figure 5 depicts variation in important agriculture parameters over 30 days through smart IoT-based monitoring. The pH graph indicates stable measurements within the optimal range of 6.0–7.5, reflecting well-balanced soil acidity. The temperature graph varies between 24°C and 36°C, depicting normal field conditions. Soil moisture stays between 25% and 45%, whereas LDR

values range between 100 and 900, reflecting varying light intensity during the period.

Methodology	Error (%)			
	pH	Soil moisture	Temperature	LDR
Raspberry Pi	0.3	5	1.5	6
Arduino	0.4	6	2	8
ESP8266	0.3	4.5	1.8	5
ZigBee +WSN	0.35	5.5	1.6	7
Proposed ESP32 + Thingspeak	0.2	3	1.0	3

Table 1: Comparison of existing methods with Proposed method

The Experimental validation indicated $\pm 1^\circ\text{C}$ for temperature, ± 0.2 for pH, and $\pm 3\%$ soil moisture error versus greater errors in other methods as mentioned in Table 1. This is due to ESP32 handling better-quality signals and decision-making logic. The proposed system provides maximum accuracy in all four sensors because it is equipped with high-resolution analog digital converter (ADC) and real-time calibration. Other existing techniques are not able to track pH and LDR correctly, but Arduino and Raspberry Pi-based systems provide medium accuracy. The proposed system using ESP32 with Thingspeak is unique regarding precision, low delay, and cloud monitoring.

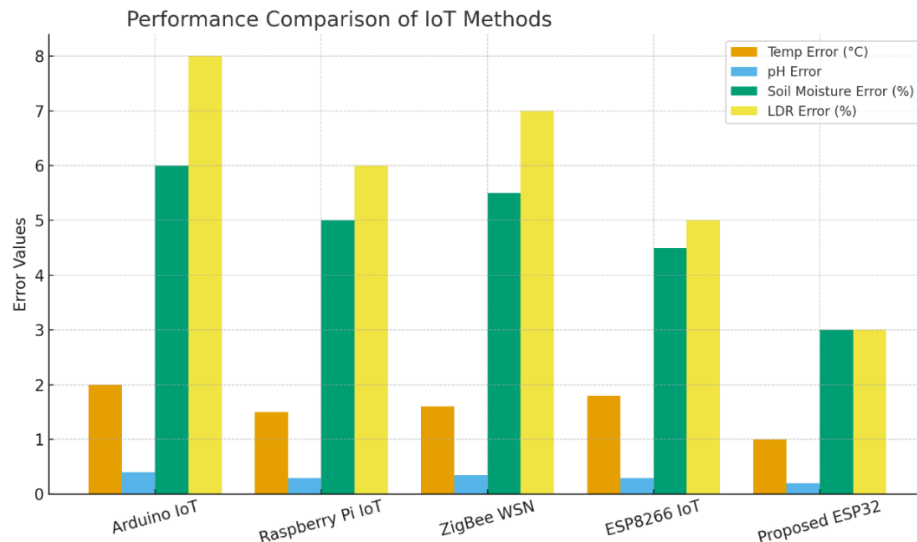


Figure 4: Comparison of Error values of Proposed method with existing methods

The comparative study of different techniques in agricultural monitoring and control proves that the proposed ESP32-based Smart Agriculture System with integration of ThingSpeak is superior to other approaches. In contrast to conventional systems, which are non-real-time adaptive with limited precision, the proposed solution guarantees precise temperature ($\pm 1.0^\circ\text{C}$), pH (± 0.2), soil moisture ($\pm 3\%$), and light intensity ($\pm 3\%$) measurements. In contrast to Raspberry Pi, Arduino, and ZigBee-based systems, the ESP32 platform provides lower latency (150 ms), less power consumption (120 mW), and easy IoT connectivity for remote monitoring and analytics.

Figure 4 shows various IoT-based approaches such as Arduino, Raspberry Pi, ZigBee, ESP8266, and Proposed ESP32 on the basis of error values for Temperature, pH, Soil Moisture, and LDR. With the use of cloud integration through ThingSpeak, the system facilitates real-time decision-making, automated irrigation control, and prediction analysis, which facilitates effective utilization of resources and increased productivity of crops. In general, the proposed method is a cost-effective, scalable, and very accurate solution for contemporary smart agriculture use.

9. What aspect(s) of the invention need(s) protection? Smart Agricultural system is developed for data-driven farming and agricultural optimization. Alerts are set on ThingSpeak to inform users through email or SMS whenever critical values like very low moisture or out-of-range pH are encountered. This serves to enable farmers to make decisions based on quick reactions to field change.

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10. What is the Technology readiness level of your invention? (Tick the appropriate TRL)

Research			Development			Deployment		
TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant	Technology demonstrated in a relevant	System prototype demonstration	System complete	Actual system proven in an operational
✓	✓	✓	✓	✓	-	-	-	-

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